	BMSCOLLEGEOF ENGINEERING,BANGALORE-19 (Autonomous Institute,AffiliatedtoVTU) **Department Name: Computer Science and Engineering**

FIRST INTERNAL

Course Code: 23CS4PCSED	**Course Title: Software Engineering**	
Semester:4th A,B,C,D,E,F	**MaximumMarks:40**	**Date:29/5/2024**
Faculty Handling the Course: Dr.LNR, SWT, RM, SB, SAK, RKM		
Instructions: Internal Choice in PART C only		

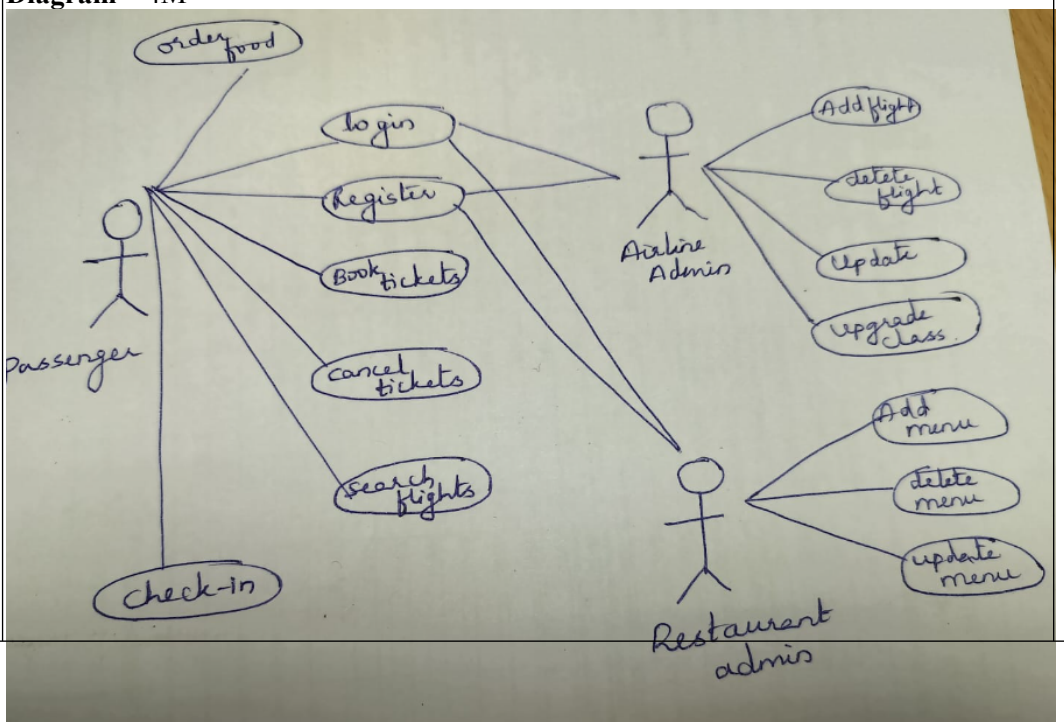
PART-A

Total 5 Marks

No.	Question	Marks	CO	PO
1	**Ethical dilemmas:** Whether to help the relative to save the relationship or to be honest to the organization.—1M Professional and Ethical Responsibilities violated: --- 2M 1. Computer misuse- Brief explanation 2. Confidentiality- Brief explanation Code of Ethics by ACM/IEEE CS --- 2M 1. PUBLIC - Software engineers shall act consistently with the public interest. 2. CLIENT AND EMPLOYER - Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest. 3. PRODUCT - Software engineers shall ensure that their products and related modifications meet the highest professional standards possible. 4. JUDGMENT - Software engineers shall maintain integrity and independence in their professional judgment. 5. MANAGEMENT - Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance. 6. PROFESSION - Software engineers shall advance the integrity and reputation of the profession consistent with the public interest. 7. COLLEAGUES - Software engineers shall be fair to and supportive of their colleagues. 8. SELF - Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.	5M	1	8

PART-B

No.	Question	Marks	CO	PO
2a	(i)Software Engineering and System Engineering --- 2M System engineering is concerned with all aspects of the development and evolution of complex systems where software plays a major role. System engineering is there fore concerned with hardware development, policy and process design and system deployment as well as software engineering. Whereas, software engineering is only concerned only with the practical problems in the development and evolution of a software. Software engineering is a part of system engineering. (ii)Exploratory development and Throw-away prototyping – 3M **Exploratory development** where the objective of the process is to work with the customer to explore their requirements and deliver a final system. The development starts with the parts of the system that are understood. The system evolves by adding new features proposed by the customer. **Throwaway prototyping** where the objective of the evolutionary development process is to understand the customer's requirements and hence develop a better requirements definition for the system. The prototype concentrates on experimenting with the customer requirements that are poorly understood.	5M	2	2

2b	Classify the following into functional, Non-functional and Domain requirements with justification for your answer. 1M each (0.5 identification and 0.5 justification) (i)The process must finish execution within three hours and data must be available by 8PM. – Non-Functional as it speaks about the time of execution and data availability (ii) User must be able to edit messages after they are sent. – Functional requirement as it speaks about the task to be performed by the user. (iii) A student to take Makeup exam should have scored Internal Marks greater that 40 marks and attendance should be above 85%. --- Domain requirement as it speaks about a specific requirement that arises from the domain of the system. (iv)Each order shall be allocated a Unique Identifier for the processing. – Functional requirement as it speaks about how the particular order needs to be processed and it is compulsory to include it. (v) The application works on all systems that has support for Windows and Linux Operating system. – Non-functional requirement as it speaks about the portability of the application.	5M	1	1
2c	Develop a set of use cases that could serve as a basis for understanding the requirements for an Airline Reservation system for fully automated operation with the help of a diagram. **Actors**: Passengers, Airline admin, restaurant admin, Bank admin/Payment Gateway representative ---- 1M **Diagram** – 4M  	5M	1	1

PART- C

Total 20 marks

No.	Question	Marks	CO	PO
3a	i) Identify and Analyze the principal viewpoints which might be taken into account and organize these using a view point hierarchy diagram for a Patient Management System in a hospital.	(6+4) 10M	2	2

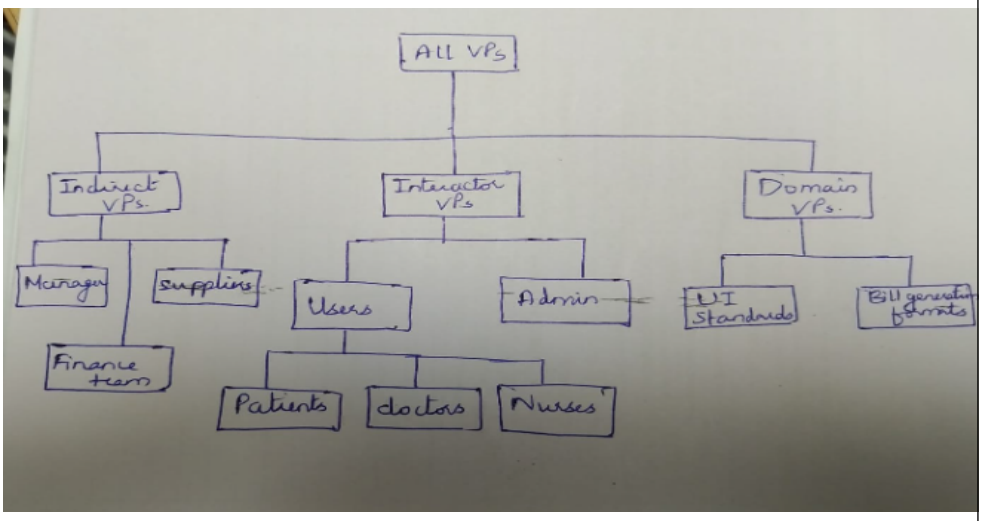


Diagram 4 M
Brief explanation of each viewpoint 2M

ii) Illustrate the different metrics used for specifying non-functional requirements

Any 4 metrics with measures – 4M

Speed	Processed transactions/second User/Event response time Screen refresh time
Size	K bytes Number of RAM chips
Ease of use	Training time Number of help frames
Reliability	Mean time to failure Probability of unavailability Rate of failure occurrence Availability
Robustness	Time to restart after failure Percentage of events causing failure Probability of data corruption on failure
Portability	Percentage of target-dependent statements Number of target systems

OR

	Design a template using structured natural language to capture the requirements for	10M	3	3

3b	a Coffee Vending Machine. Function and Description – 1M Inputs: 1M Source: 1M Outputs: 1M Action: 3M Requires, pre and post condition: 3M **Function:** Dispense coffee **Description:** This function dispenses coffee based on the options chosen by the user. **Inputs:** Options selected by the user: Hot or cold, milk or black, cappuccino or latte, current levels- milk M, coffee C, water W **Source:** The options are entered through the vending machine's interface. The current coffee level, milk level and water level taken from machine's memory. **Outputs**: The type of coffee chosen is dispensed **Destination:** Main control loop **Action:** Check the current levels of coffee, milk and water in the machine. If the level is above threshold level, start the process of making coffee. If either of them is below threshold, display an error message. The coffee is prepared by mixing the right amount of coffee (5 gms), milk and water and dispensed accordingly. The total quantity is rounded to 100 ml (70 ml of milk, 30 ml water). **Requires:** The current levels of coffee, milk and water in the machine. **Pre-**condition: The machine has the minimum amounts of ingredients required for making coffee. **Post-**condition: The current levels of coffee, milk and water are updated- M=M-70, W=W-30 and C=C-5. **Side Effects:** None.			
4a	(i)Identify the appropriate software process model for the following applications. Justify your answer a) An automated car driving system Waterfall model: Since the project aims to develop an automated car, it cant be developed in iterations or in increments. Also, we cannot start with incomplete set of requirements since any change in the requirement leads to a lot of rework which is highly expensive in case of automotive design. In waterfall approach, the next phase cannot begin until the previous phase is complete and frozen. Hence, this is the most suitable one for car driving system – 2M b) A library catalogue system that prints requested articles Component based Software Engineering (CBSE): Library system includes commonly used software modules such as login, registration, addition and deletion. These components if already developed for other projects can be reused by adapting them to the new requirements. For the requirements which are completely new we can use evolutionary model. --- 2M c) An automated robot floor cleaner – 2M Incremental Development/Evolutionary model: Since it is a robotic floor cleaner, it can be developed in increments and delivered in increments. If the requirements are not clear, we can also use throw-away prototyping to get a clearer understanding of the requirements. (ii)Identify the various problems that could arise when requirements are written in natural language. 1. **Lack of clarity:** It is sometimes difficult to use language in a precise and unambiguous way without making the document wordy and difficult to read. 2. **Requirements confusion:** Functional requirements, non-functional	(6+4) 10M	2	2

	requirements, sysicem goals and design infonnation may not be clearly distinguished. 3. **Requirements amalgamation:** Several different requirements may be expressed together as a single requirement.			
	OR			
4b	(i) Design a Software requirements document for an Online furniture ordering System as specified by IEEE/ANSI. 1. Introduction ---- 1M 1.1 Purpose of the requirements document The purpose of this document is to give a detailed description of the working of the online furniture ordering system and the constraints on its operation. 1.2 Scope of the product The product is restricted to ordering of wooden and plastic furniture items required for homes and offices. It does not allow for online payments. 1.3 Definitions, acronyms and abbreviations 1.4 References 1.5 Overview of the remainder of the document 2. General description --- 2M 2.1 Product perspective 2.2 Product Functions This product allows users to search for a specific furniture item and place an order for it to be delivered at their place. The amount is to be paid on delivery of the item. 2.3 User characteristics The users of this requirements document are clients, software developers and product managers. 2.4 General constraints 1. The product details are to be displayed both in English and Kannada. 2. The customers cannot make any type of online payments. 3. The billing address and shipping address needs to be entered compulsorily for placing an order. 2.5 Assumptions and dependencies 1. The basic assumption is that the end users have some knowledge about using the computers. 3. Specific requirements --- 3M a. Functional Requirements – (any 2) b. Non-Functional Requirements – (any 2) c. Interface Requirements 4. Appendices 5. Index	(6+4) 10M	3	3
	(ii) Identify the key challenges faced by Software Engineers in 21st Century. 3 challenges with explanation – 4M 1. Heterogeneity chellenge 2. Delivery challenge 3. Trust challenge			